

MENG 4205.002**LAB 1: IMPACT OF A JET**

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ABSTRACT

The purpose of this experiment was to study the impact of a water jet on different vane shapes and to compare the measured forces with theoretical values from momentum theory. The Impact of a Jet apparatus was used with a hydraulic bench to direct a jet of water onto a flat plate, hemispherical cup, and other vane types. The force on each vane was balanced using a jockey weight and compared with the momentum flow rate of the jet. Experimental results showed that the force increased as the mass flow rate increased linearly, and the trend was in line with theory, though minor differences were observed due to errors in measurement and non-ideal jet behavior. These errors included a 172% error for the flat vane, 46% for the cone vane, 1% for the hemisphere and lastly 131% for the angled vane. Overall, the experiment demonstrated the principle of momentum transfer in fluids and gave practical insight into how jet forces are applied in real systems such as turbines.

Keywords: Fluid mechanics, Momentum transfer, Jet impact, Hydraulics, Water turbines, Impulse turbines, Vane geometry, Conservation of momentum, Hydraulic bench, Efficiency, Force measurement, Flat plate, hemispherical cup, conical plate, angled plate

Nomenclature

F	Force exerted on vane, N
$\dot{m}$	Mass flow rate of water jet, kg/s
ρ	Density of water, kg/m^3
A	Cross-sectional area of nozzle, m^2
u	Velocity of water jet at nozzle exit, m/s
u_0	Velocity of water jet at vane (after accounting for gravity), m/s
u_1	Velocity of water after deflection, m/s
β	Deflection angle of jet from vane surface, degrees ($^\circ$)
g	Acceleration due to gravity, m/s^2
y	Distance moved by jockey weight along lever, m

s	Height of vane above nozzle tip, m
M_j	Mass of jockey weight, kg
l	Lever arm distance from vane center to pivot, m

1. INTRODUCTION

The use of water to generate power has a long history, dating back to ancient water wheels that converted the energy of flowing rivers into mechanical work for grinding grains and other tasks. Over time, this simple idea developed into more advanced hydraulic machines, eventually leading to the design of water turbines in the 19th century [1]. Pioneers like Benoît Fourneyron and James Francis laid the groundwork for modern turbines, creating designs that could harness the momentum of moving water more efficiently.

Today, several types of water turbines are in use, including Pelton, Francis, and Kaplan turbines. Each type is suited to different hydraulic conditions: Pelton turbines work best with high-head, low-flow applications; Francis' turbines are versatile and used in medium-head ranges; Kaplan turbines are designed for low-head, high-flow environments. These turbines rely on the interaction between high-velocity water jets and turbine blades to convert fluid momentum into mechanical rotation.

The efficiency of a turbine depends not only on its type but also on how effectively momentum is transferred from water to the blades. Engineers use several methods to calculate efficiency, such as comparing the input hydraulic power to the output mechanical power, analyzing velocity triangles, and applying momentum and energy equations [2]. Improvements in turbine efficiency can come from optimizing blade geometry, reducing energy losses due to turbulence and cavitation [3], and carefully controlling operating conditions. Advances in computational fluid dynamics (CFD) have also made it possible to simulate and refine designs before building physical prototypes.

Understanding how fluid jets interact with different surfaces is central to these design improvements. In this study, the Impact of a Jet apparatus is used to demonstrate the principle of momentum transfer by measuring the force exerted by a jet of

water on vanes of different shapes. This experiment provides a simplified way to connect theoretical predictions with practical observations, offering insight into the same momentum transfer principles that underpin the operation and efficiency of large-scale hydraulic turbines.

2. THEORY

The principle of this experiment is based on the conservation of momentum. A water jet striking a vane transfers momentum, generating a reaction force on the vane's surface. The force depends on the mass flow rate, jet velocity, and the angle through which the jet is deflected [4].

For a jet of mass flow rate $\dot{m}$ and velocity u_0 impinging on a vane, the change in momentum along the x-axis gives the force:

$$F = \dot{m}(u_0 - u_1 \cos \beta) \quad (1)$$

where u_1 is the exit velocity of the jet and β is the deflection angle. Under ideal conditions, the jet maintains constant velocity "isotactic" ($u_0 = u_1$), and the expression reduces to:

$$F \approx \dot{m}u_0(1 - \cos \beta) \quad (2)$$

Different vane geometries produce different deflection angles. A flat plate deflects the jet at about 90° , while a hemispherical cup reverses the flow with $\beta=180^\circ$. According to theory, the greater the change in momentum, the larger the resulting force.

The momentum flux of the jet is defined as:

$$\dot{m}u_0 = \rho A u_0^2 \quad (3)$$

where ρ is the fluid density and A is the nozzle cross-sectional area. The theoretical force for each vane geometry is compared against experimental data to evaluate how closely real jet behavior follows ideal momentum predictions.

3. METHODS

In this section, cyclic loading data of thermoplastic polyurethane from the previously reported molecular simulations is adopted to bridge with the experimental level and model nonlinear behavior for the continuum model.

3.1 Apparatus Setup and Balancing

The jet apparatus is placed on the hydraulic bench and connected to the water supply. The weight beam is set to its balanced position with the jockey weight at zero. The system is leveled using adjustable knobs, and the tally mark confirms balance before starting.

3.2 Establishing Flow and Collecting Data

Water is admitted through the supply valve to produce a vertical jet. The vane (flat plate, hemispherical cup, conical plate, or angled plate) is mounted on the lever arm. As the jet strikes the vane, the jockey weight is moved along the lever until the system returns to balance. At each flow setting, the discharge is

collected in the weighing tank up to fifteen liters over a measured time to calculate mass flow rate.

Mass flow rate ($\dot{m}$) was calculated by taking each iteration's flow rate and knowing that 1L = 1kg of water, the 15kg of water collected within the measured time of filling gave us that iteration's mass flow rate.

3.3 Repetition with Different Vanes

The procedure is repeated for each vane shape (flat plate, hemispherical cup, conical plate, angled plate). This allows comparison of how vane geometry affects the force generated.

3.4 Velocity, Force and Momentum Calculations

The jet velocity (u) is determined from the measured mass flow rate described in 2.2 and dividing it by ρA ,

$$u = \dot{m} / \rho A \text{ (m/s)} \quad (4)$$

where A is the nozzle's cross-sectional area which is given by 78.5mm^2 and ρ is the density of water (10^3 kg/m^3).

The jet velocity after deflecting the vane (u_0) can be found by acknowledging that its velocity is less than u given the deceleration of gravity and can be denoted as

$$u_0 = \sqrt{u^2 - (2 \cdot g \cdot s)} \quad (5)$$

Where g (9.81 m/s^2) is gravity's acceleration and s is the distance from the vane to the tip of the jet's nozzle.

Momentum

The theoretical force is calculated from momentum change, based on vane geometry and deflection angle.

3.5 Comparison of Experimental and Theoretical Results

Forces obtained from jockey weight balance are compared with those predicted by momentum theory. Graphs of force vs. momentum flux ($\dot{m}u_0$) are plotted for each vane. Discrepancies are analyzed in terms of jet spreading, energy losses, velocity distribution, and measurement errors.

4. EXPERIMENTS

Experiments were conducted using the Impact of a Jet apparatus mounted on a hydraulic bench. The apparatus consisted of a vertical nozzle producing a jet directed onto vanes

of different shapes, including a flat plate, hemispherical cup, conical plate, and angled plate.



Figure 1: Flat plate shown attached.

The apparatus was first leveled, and the lever arm was balanced with the jockey weight at zero position. Water flow was admitted, and the force generated on the vane was balanced by moving the jockey weight along the lever arm until equilibrium was restored. The displacement of the jockey weight was recorded for each condition.

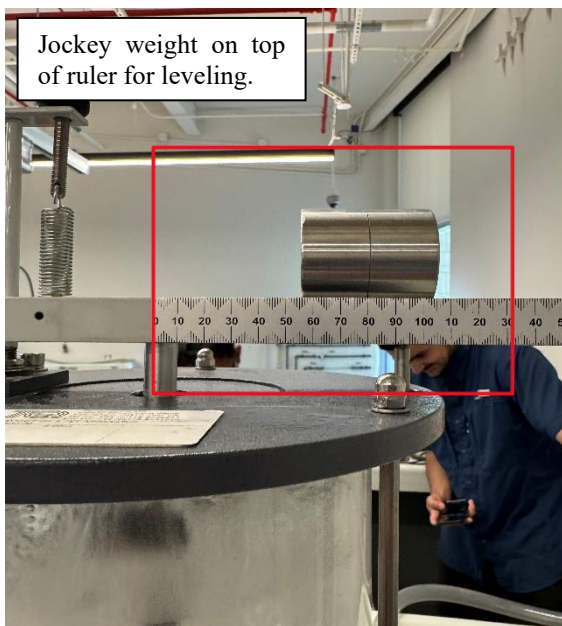


Figure 2: Jockey weight laying on top of lever arm to level the vane.

For each trial, the discharge was collected in a weighing tank for a timed interval to calculate the mass flow rate. Jet velocity at the vane was obtained from the flow data and corrected for the vertical distance between the nozzle and vane.

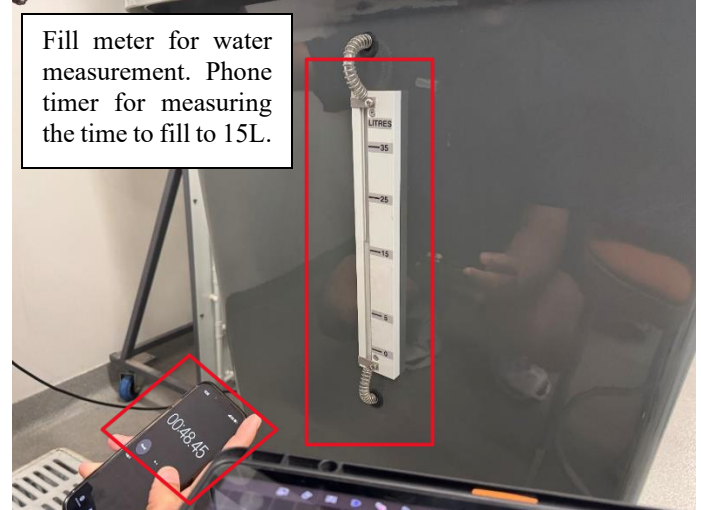


Figure 3: Discharge bin with measurement and timer shown.

The procedure was repeated for all vane geometries over a range of flow rates. The flat plate, for example, deflects the water at an angle of about 90° , while the hemispherical cup causes the water to reverse direction completely with $\beta = 180^\circ$. According to theory, the greater the change in momentum, the greater the force. The measured forces were then compared with theoretical predictions from momentum theory, and results were plotted to evaluate agreement between experimental and theoretical values.

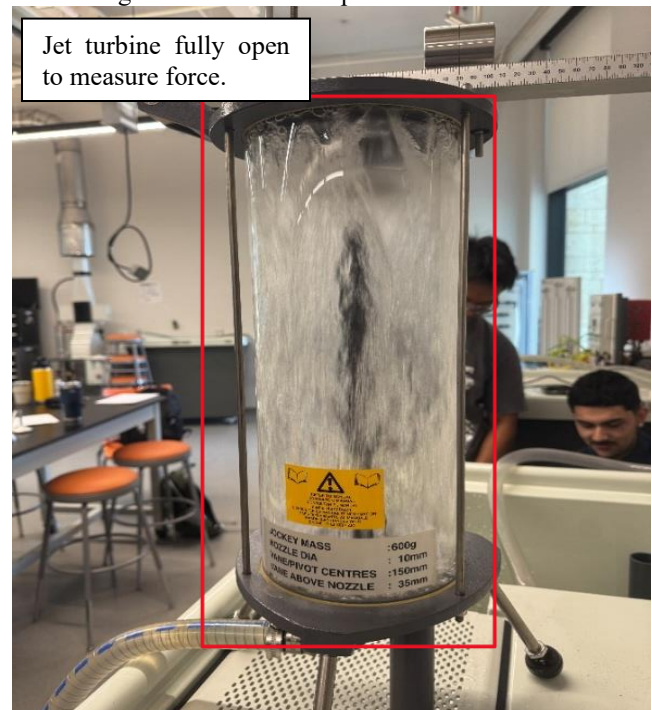


Figure 4: Jet Flow being calculated.

5. RESULTS AND DISCUSSION

5.1 Results

The experiment provided sets of measured data for each vane geometry: flat plate, conical plate, hemispherical cup, and angled plate. For each trial, the collected values of water mass, collection time, and jockey weight displacement were used to calculate the mass flow rate, jet velocity, corrected jet velocity at the vane, and both theoretical and experimental forces.

Overall, the data confirmed that the force exerted on the vane increased with mass flow rate, which is consistent with momentum theory. Among the vane geometries, the hemispherical cup produced the greatest force, while the angled plate produced the lowest. The experimental values were lower than theoretical predictions, though the trends aligned with expectations.

Quantity (kg)	t (s)	y (m)	$\dot{m}$ (kg/s)	u (m/s)	u_o (m/s)	$\dot{m}u_o$ (N)	F (N)
15.00	43.38	0.10	0.35	4.41	4.33	1.50	3.85
15.00	46.06	0.09	0.33	4.15	4.07	1.32	3.53
15.00	48.45	0.08	0.31	3.95	3.86	1.19	3.14
15.00	50.63	0.07	0.30	3.78	3.69	1.09	2.75
15.00	55.19	0.06	0.27	3.47	3.36	0.91	2.35
15.00	59.19	0.05	0.25	3.23	3.12	0.79	1.96
15.00	64.73	0.04	0.23	2.95	2.84	0.66	1.57
15.00	75.41	0.03	0.20	2.54	2.40	0.48	1.18

Table 1: Results for Flat Vane

Quantity (kg)	t (s)	y (m)	$\dot{m}$ (kg/s)	u (m/s)	u_o (m/s)	$\dot{m}u_o$ (N)	F (N)
17.00	48.22	0.15	0.35	4.50	4.42	1.56	5.93
21.00	49.47	0.15	0.42	5.41	5.35	2.27	5.85
16.50	52.89	0.13	0.31	3.98	3.89	1.21	5.14
18.50	54.91	0.12	0.34	4.30	4.21	1.42	4.71
19.00	55.67	0.11	0.34	4.35	4.27	1.46	4.32
17.50	56.65	0.10	0.31	3.94	3.85	1.19	3.92
18.50	60.57	0.09	0.31	3.89	3.81	1.16	3.53
17.50	63.75	0.08	0.27	3.50	3.40	0.93	3.14
16.50	65.94	0.07	0.25	3.19	3.08	0.77	2.75

Table 2: Results from Conical Vane

Quantity (kg)	t (s)	y (m)	$\dot{m}$ (kg/s)	u (m/s)	u_o (m/s)	$\dot{m}u_o$ (N)	F (N)
17.00	48.22	0.15	0.35	4.50	4.42	1.56	5.89
21.00	49.47	0.15	0.42	5.41	5.35	2.27	5.89
16.50	52.89	0.13	0.31	3.98	3.89	1.21	5.10
18.50	54.91	0.12	0.34	4.30	4.21	1.42	4.71
19.00	55.67	0.11	0.34	4.35	4.27	1.46	4.32
17.50	56.65	0.10	0.31	3.94	3.85	1.19	3.92
18.50	60.57	0.09	0.31	3.89	3.81	1.16	3.53
17.50	63.57	0.08	0.28	3.51	3.41	0.94	3.14

Table 3: Results from Hemispherical Vane

Quantity (kg)	t (s)	y (m)	$\dot{m}$ (kg/s)	u (m/s)	u_o (m/s)	$\dot{m}u_o$ (N)	F (N)
17.00	48.22	0.15	0.35	4.50	4.42	1.56	5.89
21.00	49.47	0.15	0.42	5.41	5.35	2.27	5.89
16.50	52.89	0.13	0.31	3.98	3.89	1.21	5.10
18.50	54.91	0.12	0.34	4.30	4.21	1.42	4.71
19.00	55.67	0.11	0.34	4.35	4.27	1.46	4.32
17.50	56.65	0.10	0.31	3.94	3.85	1.19	3.92
18.50	60.57	0.09	0.31	3.89	3.81	1.16	3.53
17.50	63.57	0.08	0.28	3.51	3.41	0.94	3.14

Table 41: Results from Angled Vane

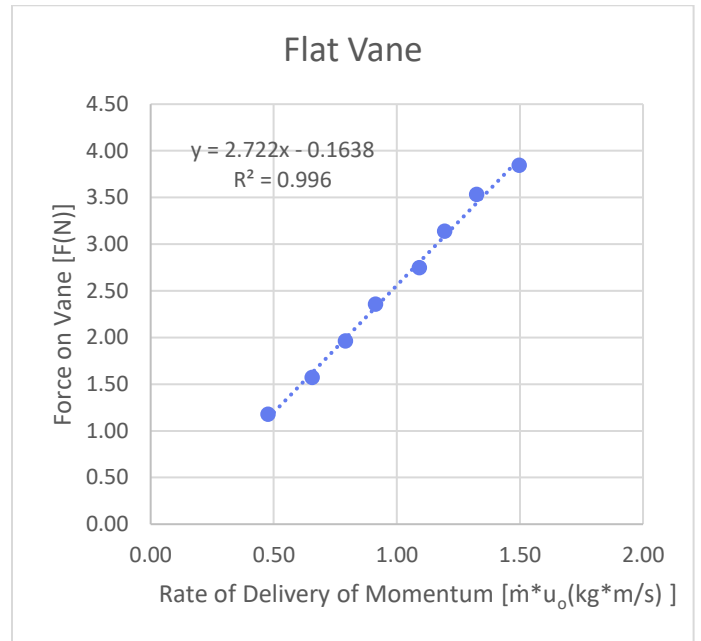


Figure 5 2: Graphical representation of Force & Delivery Momentum for Flat vane with a slope of 2.722

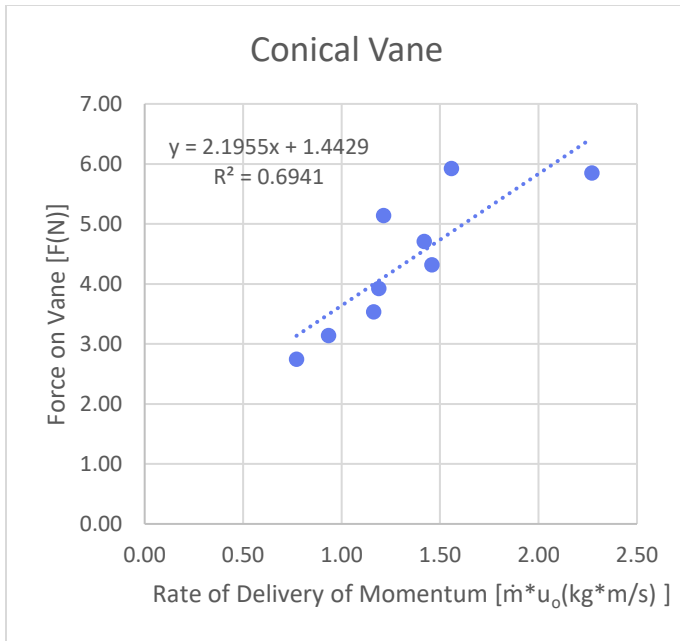


Figure 63: Graphical Representation of Force and Delivery Momentum for Conical Vane with a slope of 2.1955

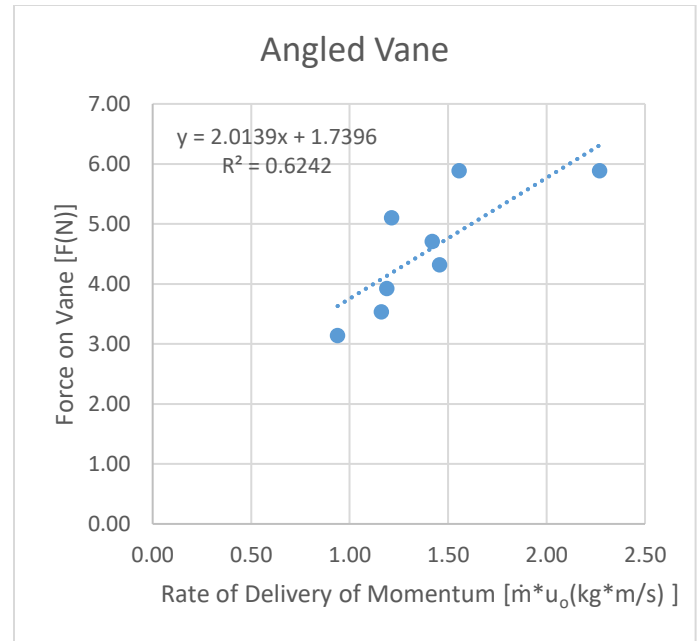


Figure 84: Graphical Representation of Force and Delivery Momentum for Hemispherical Vane with a slope of 2.014

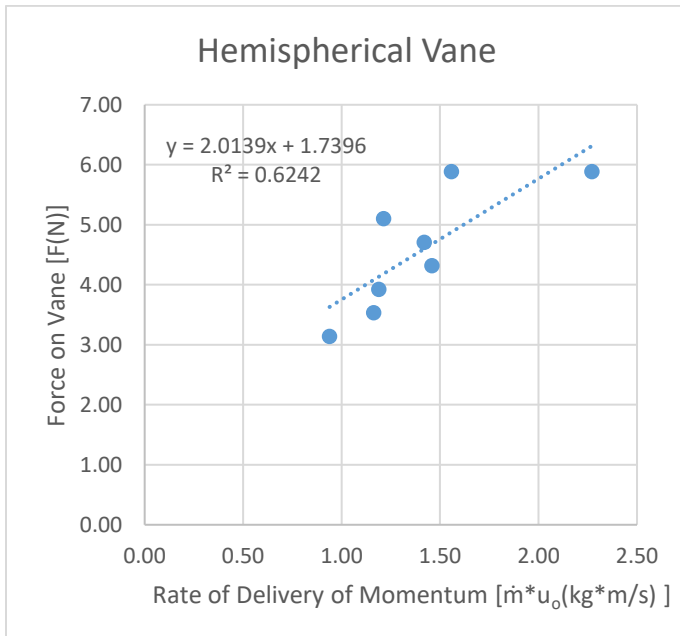


Figure 7: Graphical Representation of Force and Delivery Momentum for Angled Vane with a slope of 2.014

5.2 Sample Calculations

Measured: Collected mass = 5.4 kg in 60 s $\rightarrow \dot{m} = 0.09$ kg/s

Jet exit velocity:

$$u = \dot{m} / (\rho A) = 0.0900 / (1000 \times 7.85 \times 10^{-5}) = 1.147 \text{ m/s}$$

Velocity at vane:

$$u_0^2 = u^2 - 2gs = (1.147)^2 - 2(9.81)(0.035) = 0.6283 \rightarrow u_0 = 0.794 \text{ m/s}$$

Momentum flux:

$$\dot{m}u_0 = 0.0900 \times 0.794 = 0.0715 \text{ N}$$

Theoretical force:

$$\begin{aligned} \text{Flat} &= 0.0715 \text{ N} \\ \text{Conical} &= 0.107 \text{ N} \\ \text{Hemisphere} &= 0.143 \text{ N} \\ 30^\circ \text{ angled} &= 0.062 \text{ N} \end{aligned}$$

Experimental force ($y = 3.00$ mm = 0.00300 m):

$$F = 4gy = 4(9.81)(0.00300) = 0.118 \text{ N}$$

Comparison (Conical):

$$\begin{aligned} \text{Theoretical} &= 0.107 \text{ N} \\ \text{Experimental} &= 0.118 \text{ N} \\ \rightarrow \text{Error} &\approx 10.3\% \end{aligned}$$

5.3 Ideal vs. Actual Comparison

Shapes	Ideal Slopes	Actual Slopes	% Error
Flat Plate	1	2.722	172%
Cone	1.5	2.196	46%
Hemisphere	2	2.014	1%
Angle	0.87	2.014	131%

Table 5 Shows the results from our actual vs the ideal calculations.

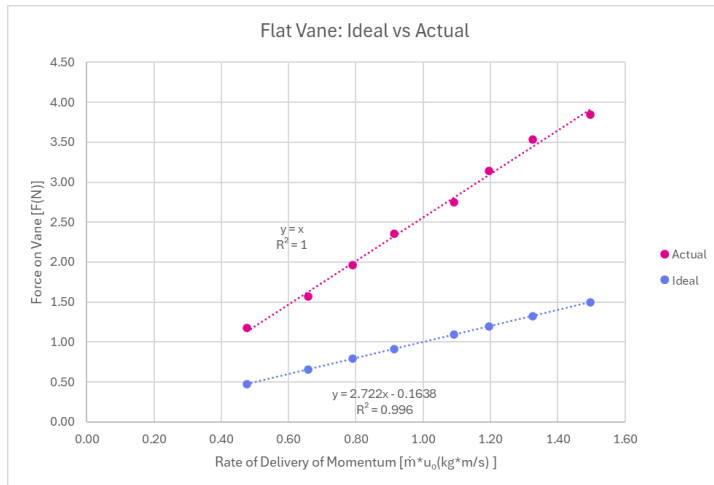


Figure 9: Graph showing Ideal vs Actual trendline for Flat Vane

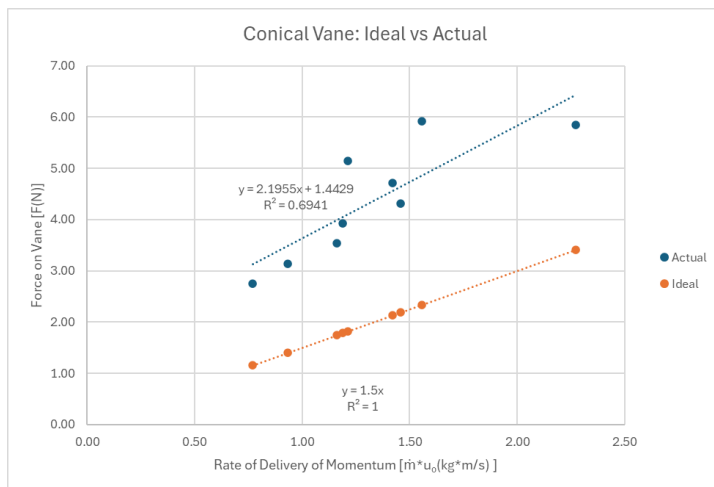


Figure 10: Graph showing Ideal vs Actual trendlines for Conical Vane

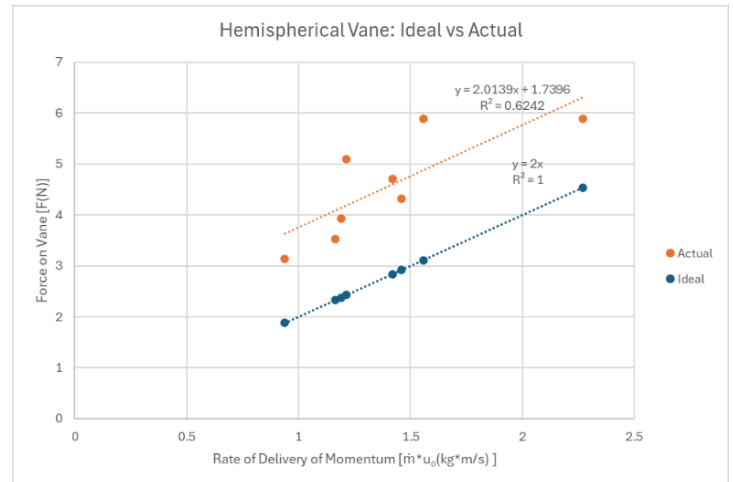


Figure 11: Graph showing Ideal vs Actual trendline for Hemispherical Vane

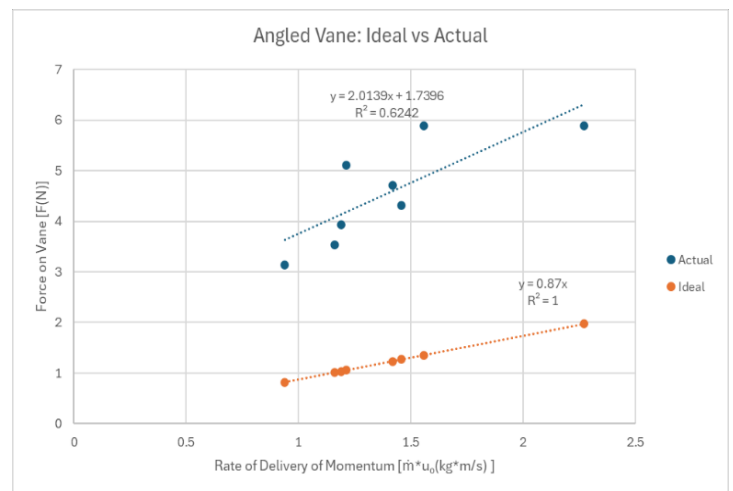


Figure 12: Graph showing Ideal vs Actual trendline for Angled Vane

5.4 Discussion

The results clearly showed that the force on each vane increased with mass rate, in line with momentum theory. The hemispherical cup generated the highest force since it reversed the jet completely, while the angled plate produced the least because it redirected the jet by only a small angle. The flat and conical plates fell in between, consistent with the expected momentum changes.

Although the experimental trends matched theory, the measured forces were lower than predicted. Part of this difference comes from the fact that a real jet does not have a uniform velocity profile. Instead, velocity is highest at the center and decreases toward the edges, which reduces the effective momentum transfer. The jet also spreads and loses speed as it travels from the nozzle to the vane. Even a modest change, say, the jet area increasing by about ten percent while the velocity decreases by the same amount would reduce the effective momentum flow rate and, in turn, lower the measured force by around ten percent compared to the ideal case.

Some small systematic errors played a role as well. If the jockey weight's actual mass were slightly off or the lever arm distance from pivot to vane were mismeasured by even a millimeter, every calculated force would shift a little higher or lower. These errors do not change the overall trends, but they do affect efficiency calculations, making the experimental values consistently deviate from theory. A minor "zero error" was also noticeable at times when the lever balance was not perfect, giving the impression of force even with no jet applied.

Another factor is that the current apparatus relies on manual balancing and water collection, which introduces room for human error. Using a digital force sensor in place of the jockey weight or an electronic flow meter instead of manual timing would improve precision and reduce the scatter in the data.

Despite these limitations, the experiment still succeeded in demonstrating the key principle: force is proportional to the momentum change of the jet. It also highlighted the importance of vane geometry in fluid machinery, where blade shape has a direct effect on efficiency. In real systems like turbines, careful design ensures that flow is captured and redirected effectively. For example, if the cone or hemisphere were reversed so that the jet hit their convex surfaces, the water would mostly splash and flow around them rather than being redirected. In that case, momentum theory would predict almost no force, but in practice turbulent separation and splashing would create unpredictable forces. This gap between simple theory and messy reality illustrates why engineers must account for real fluid behavior when designing systems.

6 CONCLUSION

This experiment demonstrated how the force exerted by a water jet depends on both the flow rate and the geometry of the vane it strikes. As the mass flow rate increased, the measured force on every vane increased proportionally, confirming the general trend predicted by momentum theory. The hemispherical cup produced the greatest force because it redirected the jet 180° , while the angled plate produced the least due to its small deflection angle.

Although the experimental results were larger than theoretical values, the differences were expected. Real jets spread, slow down and lose velocity between the nozzle and the vane, and the velocity distribution across the jet is not perfectly uniform. Small measurement errors, such as balancing the lever arm or timing water collection, also introduced some variation. Even with these limitations, the results followed the predicted trends and clearly demonstrated the relationship between jet momentum and force.

Beyond validating momentum theory, the experiment underscored the importance of vane shape in fluid machinery. Just as the hemispherical cup captured more momentum than the flat plate, turbine blades and other hydraulic components must be carefully designed to maximize efficiency. The work also highlighted practical considerations: improving instrumentation with digital sensors or electronic flow meters could reduce

human error and bring measured values closer to theoretical predictions.

Overall, the experiment successfully linked theory to practice, providing insight into both the strengths and the limitations of applying ideal fluid assumptions to real-world systems.

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